

ORF 245: Fundamentals of Statistics – Spring 2025

Precept 1

Last updated: January 24, 2025

1. (Ice breaking + Introductions)

- Introduce yourself.
- Set communication ground rules: e.g., can stick around for up to 10 minutes after precept. Additional help from office hours.

2. (R Introduction)

- Idea here is to introduce some basics to get them started. Here is a suggested outline (Rmd file ‘R-intro.Rmd’ in the Supplemental folder). I also include another quick R tutorial in the file ‘Supplemental/R-tutorial.pdf’.
- Introduction of R-Studio, how to write notes and codes using R-Markdown, run code in a chunk, `plot`, visualization in R-markdown.
- Some basics of R
 - R as a calculator.
 - vector
 - * Create a vector using function `c(...)`.
 - * Vector operation: elementwise plus/multiplication, scalar multiplication
 - * Vector generator: `rep`, `seq`
 - * The help command e.g. `?rep`
 - * Track the value of vector in ‘Environment’.
 - matrix
 - * Create a matrix using function `matrix`.
 - * Matrix operation: elementwise plus/multiplication, matrix multiplication (`%*%`), transpose
 - * Function `nrow`, `ncol`, `length`
 - * Concatenate two matrices in row (`rbind`) or in column (`cbind`).
 - data.frame (short introduction, use `?data.frame` for details).
- R as a programming language
 - function
 - * `print`
 - * summary statistics: `mean`, `median`, `min`, `max`, `sum`.
 - if-then-else `if (...) { ... } else { ... }`.
 - for-loop `for (i in 1 : 10) {...}`.
 - data types: numeric value, int, logical values (`TRUE/FALSE`), string/characters, complex
 - logical statements, e.g. not equal to `!=`, or `||`, and `&&`.
- Data manipulation
 - subsetting (see examples in the Rmd file), the logits behind `x[x < 0.3]`.
 - encourages the use of vector operation for fast computation.